

Diffie–Hellman Key Exchange Algorithm

Aim

To implement the **Diffie–Hellman Key Exchange Algorithm** for securely generating a shared secret key between two communicating parties over an insecure communication channel.

Algorithm

Step 1: Start the program.

Step 2: Select a large prime number p and a primitive root g .

Step 3: User A chooses a private key a .

Step 4: User B chooses a private key b .

Step 5: User A computes the public key. $A = g^a p$

Step 6: User B computes the public key. $B = g^b p$

Step 7: Exchange the public keys.

Step 8: User A computes the shared secret key. $K = B^a$

Step 9: User B computes the shared secret key. $K = A^b p$

Step 10: Both users obtain the same shared secret key.

Step 11: Stop the program.

C Program

```
#include <stdio.h>

long long power(long long base, long long exp, long long mod)
{
    long long result = 1;

    while(exp > 0)
    {
        if(exp % 2 == 1)
            result = (result * base) % mod;
        base = (base * base) % mod;
        exp = exp / 2;}
    return result;}

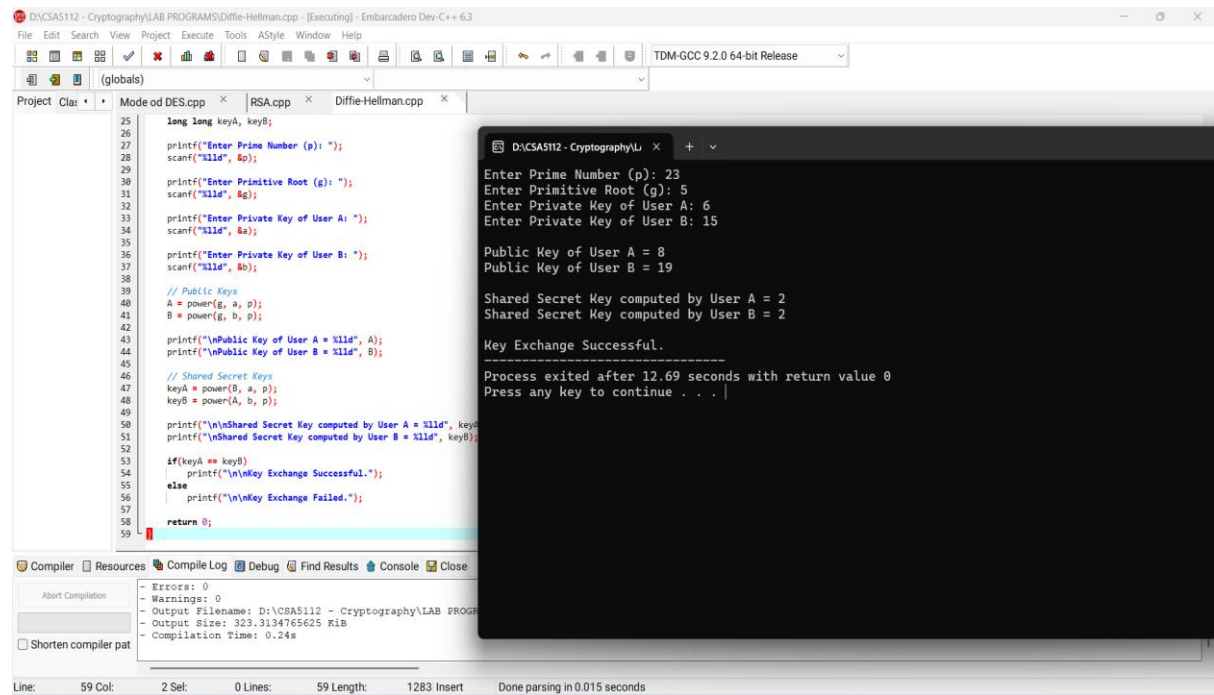
int main()
{
```

```

long long p, g;
long long a, b;
long long A, B;
long long keyA, keyB;
printf("Enter Prime Number (p): ");
scanf("%lld", &p);
printf("Enter Primitive Root (g): ");
scanf("%lld", &g);
printf("Enter Private Key of User A: ");
scanf("%lld", &a);
printf("Enter Private Key of User B: ");
scanf("%lld", &b);
A = power(g, a, p);
B = power(g, b, p);
printf("\nPublic Key of User A = %lld", A);
printf("\nPublic Key of User B = %lld", B);
keyA = power(B, a, p);
keyB = power(A, b, p);
printf("\n\nShared Secret Key computed by User A = %lld", keyA);
printf("\n\nShared Secret Key computed by User B = %lld", keyB);
if(keyA == keyB)
    printf("\n\nKey Exchange Successful.");
else
    printf("\n\nKey Exchange Failed.");
return 0;
}

```

OUTPUT:



```
D:\CSA5112 - Cryptography\LAB PROGRAMS\Diffie-Hellman.cpp - [Executing] - Embarcadero Dev-C++ 6.3
File Edit Search View Project Execute Tools AStyle Window Help
(globals)
Project Claz Mode od DES.cpp x RSA.cpp x Diffie-Hellman.cpp x
25 long long keyA, keyB;
26
27 printf("Enter Prime Number (p): ");
28 scanf("%lld", &p);
29
30 printf("Enter Primitive Root (g): ");
31 scanf("%lld", &g);
32
33 printf("Enter Private Key of User A: ");
34 scanf("%lld", &a);
35
36 printf("Enter Private Key of User B: ");
37 scanf("%lld", &b);
38
39 // Public Keys
40 A = power(g, a, p);
41 B = power(g, b, p);
42
43 printf("\nPublic Key of User A = %lld", A);
44 printf("\nPublic Key of User B = %lld", B);
45
46 // Shared Secret Keys
47 keyA = power(B, a, p);
48 keyB = power(A, b, p);
49
50 printf("\n\nShared Secret Key computed by User A = %lld", keyA);
51 printf("\n\nShared Secret Key computed by User B = %lld", keyB);
52
53 if(keyA == keyB)
54 | printf("\n\nKey Exchange Successful.");
55 else
56 | printf("\n\nKey Exchange Failed.");
57
58 return 0;
59

D:\CSA5112 - Cryptography\LAB PROGRAMS\Diffie-Hellman.cpp
Enter Prime Number (p): 23
Enter Primitive Root (g): 5
Enter Private Key of User A: 6
Enter Private Key of User B: 15

Public Key of User A = 8
Public Key of User B = 19

Shared Secret Key computed by User A = 2
Shared Secret Key computed by User B = 2

Key Exchange Successful.

Process exited after 12.69 seconds with return value 0
Press any key to continue . . . |

Compiler Resources Compile Log Debug Find Results Console Close
Abort Compilation
- Errors: 0
- Warnings: 0
- Output Filename: D:\CSA5112 - Cryptography\LAB PROGRAMS\Diffie-Hellman.exe
- Output Size: 323.3134765625 KiB
- Compilation Time: 0.24s
Shorten compiler path
Line: 59 Col: 2 Sel: 0 Lines: 59 Length: 1283 Insert Done parsing in 0.015 seconds
```